A Novel Dynamic Warping Fusion (DWF) Method for Full-Waveform Hyperspectral LiDAR Signal Correction and Decomposition

Yishuo Hao, *Student Member, IEEE*, Zheng Niu, Wang Li, Jie Bai, Li Wang, *Member, IEEE*, and Kaiyi Bi

***Abstract*—Full-waveform hyperspectral LiDAR (HSL) provides multichannel echo signals that capture both geometric and spectral information. However, the instrument's optical system introduces challenges like interchannel time-delay effects and multichannel response heterogeneity, leading to uncertainties in HSL waveform parameter extraction and quantitative applications. Therefore, we propose a dynamic warping fusion (DWF) method for HSL signal correction and decomposition. The DWF method incorporates derivative dynamic time warping (DDTW) for multichannel signal alignment and panchromatic band waveform synthesis, eliminating uncertainties in target position extraction. Subsequently, inverse DDTW is applied to generate unique initialization parameters for each channel, reducing the risks of missed detection and oversegmentation. The results suggest that: 1) the DWF method demonstrates excellent decomposition performance on simulated datasets, exhibiting robustness across varying interchannel time-delay levels (R^2 > 0.97 and root-mean-square error (RMSE) <0.24); 2) the synthesized panchromatic band demonstrates superior spatial information extraction capabilities, with an average relative neighbor distance error (RNDE) <2.51%; 3) a new multiband LiDAR spectral similarity index (MLSI) is introduced, showing that spectral curves based on waveform integrated areas are closer to the true spectrum than those based on peak intensities; and 4) performance validation on measured datasets confirms the superiority of DWF over the previous multichannel interconnection waveform decomposition (MIWD) approach, reducing RNDE from 5.11% to 2.27%. The proposed approach leverages waveform shape features for interchannel time-delay correction and waveform decomposition, providing a valuable reference for signal decomposition across various HSL devices and similar data processing tasks.**

***Index Terms*—Full-waveform light detection and ranging (LiDAR), hyperspectral LiDAR (HSL), signal alignment, waveform decomposition.**

I. Introduction

FULL-WAVEFORM light detection and ranging (LiDAR) has gained widespread application in the field of Earth observation in recent years, as it provides more abundant vertical information than passive remote sensing and more detailed echo energy distribution than discrete LiDAR [1], [2], [3], [4]. Benefiting from its ability to accurately describe the geometric characteristics of targets, full-waveform LiDAR technology has been used in various scientific research and engineering applications, such as forest structure inversion and aboveground biomass estimation [5], [6], [7], [8], [9], tree species classification and biodiversity mapping [10], [11], [12], [13], [14], estimation of plant biochemical and structural parameters [15], [16], [17], land cover classification [18], [19], [20], [21], water depth and topography measurement [22], [23], [24], as well as archeology and cultural heritage conservation [25], [26], [27].

To compensate for the spectral detection limitations of LiDAR systems using single-wavelength laser sources, significant efforts have been made to develop multispectral/hyperspectral LiDAR (MSL/HSL) systems [28], [29], [30], [31], [32], [33]. Traditional passive hyperspectral remote sensing measures the spectral radiance from targets, which results from the interaction between incident solar radiation (direct and sky diffuse components) and surface materials, as well as thermal emission from the targets [34], [35], [36]. The broadband spectral characteristics of sunlight in the visible-to-near-infrared region provide the physical foundation for hyperspectral detection. In recent years, with the development of supercontinuum laser sources, it has become possible to achieve active hyperspectral detection through direct emission of broadband laser radiation by LiDAR systems. When narrowband pulsed laser pumps nonlinear media such as photonic crystal fiber (PCF), the laser spectrum can be significantly broadened under the influence of nonlinear effects such as self-phase modulation and four-wave mixing [37], [38], [39], [40], [41]. Wallace et al. [42] designed a four-channel MSL

Received 28 January 2025; revised 13 June 2025 and 21 August 2025; accepted 5 November 2025. Date of publication 10 November 2025; date of current version 21 November 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3901305, in part by the Independent Deployment Project of Aerospace Information Research Institute of Chinese Academy of Sciences under Grant E4Z202021F, in part by the Open Fund of State Key Laboratory of Remote Sensing and Digital Earth under Grant OFSLRSS202516, in part by the National Natural Science Foundation of China under Grant 42522106 and Grant 42501448, and in part by the Intergovernmental International Science and Technology Innovation Cooperation Program through the National Key Research and Development Plan under Grant 2024YFE0198600. *(Corresponding authors: Kaiyi Bi; Li Wang.)*

Yishuo Hao and Zheng Niu are with the State Key Laboratory of Remote Sensing and Digital Earth, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: haoyishuo23@mails.ucas.ac.cn; niuzheng@radi.ac.cn).

Wang Li, Li Wang, and Kaiyi Bi are with the State Key Laboratory of Remote Sensing and Digital Earth, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China (e-mail: liwang@radi.ac.cn; wangli@aircas.ac.cn; biky@radi.ac.cn).

Jie Bai is with the School of Resources and Environment, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: baijie@uestc.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3631229, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3631229

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

system based on the acousto-optic tunable filter (AOTF) and applied it to restore structural and physiological information of coniferous forests. Chen et al. [43], [44] and Shao et al. [32], [45] developed a 91-channel HSL with a spectral resolution of 5 nm for rock classification and vegetation detection. Salido-Monzú and Wieser [46] and Han et al. [47], [48] conducted research on polarization HSL and assessed its potential in detecting and classifying material physical properties. Song et al. [49] developed an HSL system with long-range detection capabilities and explored waveform decomposition and weak information extraction algorithms. The basic technical path of HSL hardware integration involves using a supercontinuum spectral laser source to emit broad-spectrum laser pulses, receiving the target's backscattered echo signal, dispersing it through a grating into multiple spectral channels, and ultimately converting each channel's signal into digitized time series waveforms [50]. Hyperspectral waveforms can effectively characterize the spatial structure and spectral reflectance of targets within the laser footprint [51], [52], [53], but they also possess more complex data features, which introduce additional challenges for signal decomposition [54], [55].

Consequently, conventional single-wavelength decomposition methods like Gaussian decomposition (GD) [3], [56] and waveform deconvolution [57], [58], [59] may struggle to adequately decompose hyperspectral waveforms and fully exploit their rich information. With increasing attention to hyperspectral waveform decomposition, several effective algorithms have been proposed. For instance, Song et al. [49] proposed a multispectral waveform decomposition (MSWD) method, which is based on the core idea of sorting signal component energies across multiple channels to discover hidden weak and overlapping signals, demonstrating excellent performance in extracting comprehensive initial parameters. Bai et al. [50], [54] developed a ranking central locations of natural target echoes (Rclonte) method, which leverages multichannel data to estimate signal positions more precisely, excelling in ranging accuracy and spectral extraction of natural targets. However, due to factors such as optical path differences within HSL system internal components and nonsynchronous photoelectric conversion responses, hyperspectral echo waveforms often exhibit pronounced interchannel time delays [54], [55]. The quality of HSL signals is often compromised by the low reflectance of natural targets in certain spectral regions and the nonuniform energy distribution of the supercontinuum laser source, resulting in signal attenuation or overlap [52], [60]. In response to these problems, Wang et al. [55] proposed a multichannel interconnection waveform decomposition (MIWD) method, which corrects, selects, and fuses multichannel signals to improve waveform decomposition accuracy. Nevertheless, its signal correction in the time domain relies on the inherent optical path differences of internal hardware such as optical fibers. This method theoretically requires individual calibration for each instrument, and factors like internal component vibration and aging could potentially affect the optical path differences between channels, thus limiting the universality of the hyperspectral waveform decomposition method.

At the current stage, further research is still needed on how to better correct interchannel time-delay effects in HSL and improve the consistency of Gaussian component numbers and positions across channels. To explore more universal HSL signal processing algorithms, it is essential to investigate the inherent features of multichannel waveforms. Since the outgoing laser is emitted in the form of supercontinuum spectrum, signals from all channels represent the same target components within the laser footprint [52], [61]. Therefore, waveforms across all channels should exhibit identical numbers and positions of peaks, despite variations in target reflectance at different wavelengths. For curves with similar characteristics but temporal misalignment, dynamic time warping (DTW) provides a robust method for similarity measurement and alignment [62], [63]. DTW and its variants have demonstrated excellent performance across various fields, including speech recognition [63], biomedicine [64], [65], [66], meteorology [67], [68], [69], time series classification [70], [71], [72], land cover mapping [73], and crop classification [74], [75]. Given that HSL echo signals are essentially similar time series, developing DTW-based algorithms for HSL multichannel waveform correction and decomposition is promising but has yet to be explored.

In this article, we introduce the latest second-generation HSL (HSL-II) developed by our laboratory. We propose a novel dynamic warping fusion (DWF) method that can be used for robust hyperspectral signal waveform decomposition at different interchannel time-delay levels. The main objectives of this article are as follows.

1) Propose a new DWF method for aligning, fusing, and decomposing HSL signal waveforms from different channels.
2) Assess the robustness of waveform alignment at different interchannel time-delay levels, and analyze the ranging accuracy and spectral extraction capabilities.
3) Verify the waveform decomposition performance of the DWF method using simulated and measured data, and compare it with the GD and MIWD methods.

II. Materials

A. Second-Generation HSL System

The instrument used in this study is the latest HSL-II system developed by our laboratory, as shown in Fig. 1. Compared to the previous generation [16], [33], [50], the HSL-II employs a compact PXI Express (PXIe) chassis, featuring reduced size, enhanced data acquisition speed, and improved system flexibility. The modular design enables convenient component replacement and system reconfiguration to meet diverse experimental requirements.

The system consists of four main components as follows.

1) Laser Source: A supercontinuum pulsed laser generating broadband emission from 400 to 2400 nm, with a 20-kHz repetition frequency, approximately 2-ns full-width at half-maximum (FWHM), and 110-mW total average power.

2) Telescope and Scanning Platform: This subsystem collimates the laser beam and controls telescope orientation for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

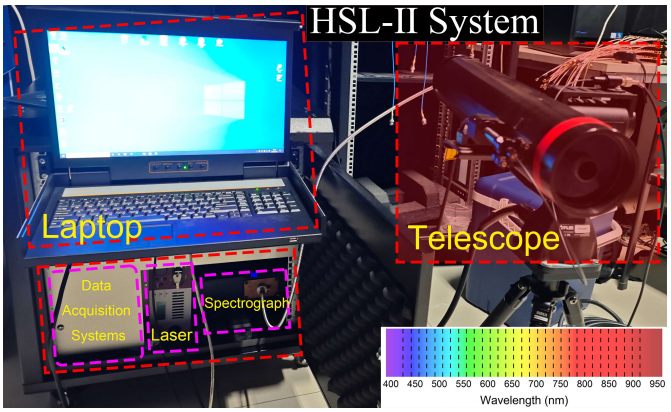

Fig. 1. Photograph of the HSL-II, showing the instrument's main components. The dashed line in the bottom right indicates the positions of 31 spectral bands.

TABLE I
PARAMETER VALUES USED IN GENERATING THE SIMULATED DATA

Signal components	Distance (m)	Central location (samples)	FWHM (samples)	Target object
Gaussian_1	8.69	290	22	petal
Gaussian_2	9.59	320	18	green leaf
Gaussian_3	11.09	370	20	SRP
Gaussian_4	12.59	420	25	stone
Gaussian_5	13.49	450	30	plywood

precise target illumination and backscatter collection. The telescope features an 80-mm optical aperture and approximately 3-mrad laser divergence angle.

3) Spectral Splitting and Photoelectric Conversion: A customized diffraction grating disperses the returned signal into 31 spectral bands spanning 425–914 nm, with 16–17-nm spectral resolution. Photomultiplier tubes (PMTs) are used for subsequent photoelectric conversion.

4) Data Acquisition and Control Center: A 5-GHz data acquisition card samples waveforms at 0.2-ns intervals, enabling full-waveform capture across all spectral bands. The entire system operation is controlled via a laptop computer.

B. Generation of Simulated Data

To evaluate the theoretical feasibility and applicability of the DWF method, simulated full-waveform HSL echo signals were generated using Python 3.7. These simulated signals consist of intensity–time curves spanning 31 spectral bands, corresponding to the HSL-II instrument. To ensure consistency with measured data, Gaussian noise with a standard deviation value of 0.2 mV and a sampling interval of 0.2 ns was applied.

Each spectral band comprises five signal components, representing different targets: petal, green leaf, 99% standard reflectance panel (SRP), stone, and plywood. The spectral characteristics of these targets were derived from the USGS Spectral Library (https://crustal.usgs.gov/speclab/QueryAll07a.php, last accessed on October 12, 2024). The echo intensities of all targets were calculated based on their reflectance values. The remaining signal parameters are detailed in Table I. To validate the DWF method's adaptability to interchannel time-delay effects, three delay scenarios were simulated by introducing random temporal offsets within ranges of [−10, 10], [−20, 20], and [−30, 30] samples for each spectral band.

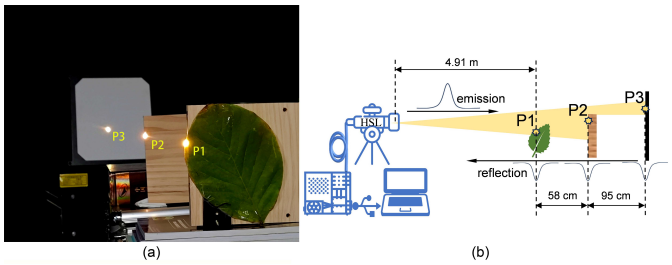

Fig. 2. (a) Photograph and (b) schematic of the measurement configuration.

C. Design of Measurement Experiment

In a laboratory with LED indoor lighting, measurements were conducted using the HSL-II on a three-target scene. The targets were arranged in sequence from the laser source: a leaf at 4.91 m from the source, followed by plank at 5.49 m, and an SRP with 99% reflectance at 6.44 m. The true positions of all the targets are obtained by a tape measure. The telescope was adjusted to ensure vertical laser illumination at the target edges, creating three subspots and recording their superimposed waveforms, as shown in Fig. 2. The supercontinuum laser emits 20 000 laser pulses per second, and the waveform collected for each laser footprint is obtained by averaging the echo signals of 30 repeated laser pulses. To evaluate the spectral extraction performance of subspots in subsequent experiments, full-spot echo signals from all three targets were also recorded.

III. METHODS

The interchannel time-delay effects and weak responses of specific bands introduce considerable uncertainty to HSL waveform decomposition. To address these challenges, we propose a DWF method that aligns, integrates, and decomposes HSL echo signals by utilizing multiband waveform features.

The DWF method encompasses four sequential steps: 1) HSL signal modeling and preprocessing; 2) waveform alignment and fusion via derivative DTW (DDTW); 3) panchromatic band autonomous GD (AGD); and 4) all channels parameter estimation via inverse DDTW. Fig. 3 provides a detailed illustration of this methodological workflow. We use the simulated and measured datasets for the performance evaluation of DWF and compare it with the GD and MIWD methods.

A. HSL Signal Modeling and Preprocessing

The basis of HSL signal decomposition is feature analysis and modeling of waveforms from multiple channels. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

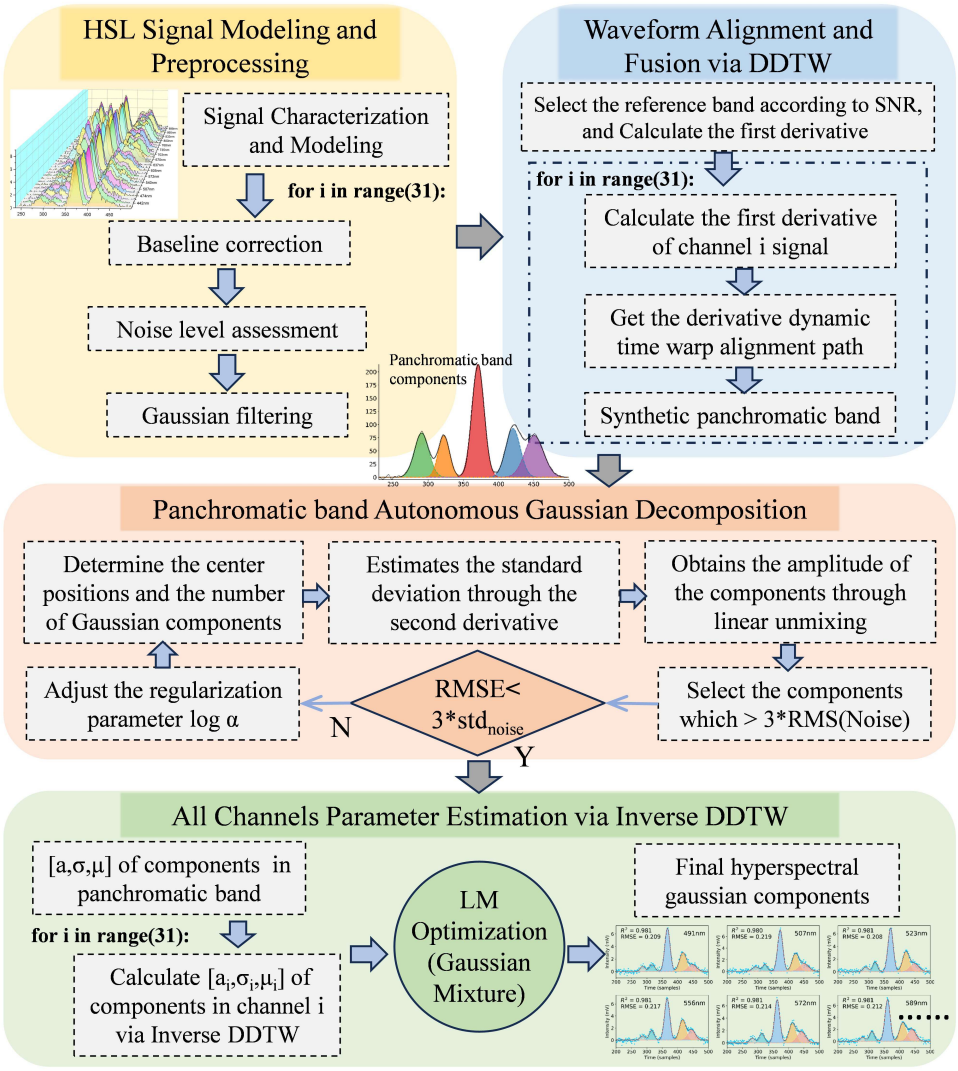

Fig. 3. Workflow of the DWF method proposed in this study.

backscattered echo signal received by LiDAR is the result of laser pulse interactions with targets within each laser footprint, which can be conceptualized as a superposition of multiple Gaussian functions [3], [54], [56], [76], [77], [78], expressed as

$$f(t) = \sum_{i=1}^{n} \frac{a_i}{\sqrt{2\pi}\sigma_i} \exp \frac{-(t-\mu_i)^2}{2\sigma_i^2} \tag{1}$$

where n represents the number of Gaussian components; a_i, σ_i, and μ_i are the amplitude, standard deviation, and central position of the ith Gaussian component, respectively; and $f(t)$ is the fitting function of the echo signal.

The HSL system also follows this principle, but with the complexity of multiple wavelength channels, where the signal in each channel is influenced by the spectral characteristics of the target in the corresponding spectral band. Moreover, full-width at half-maximum (FWHM) is considered a critical parameter for assessing target geometric and structural characteristics [78], [79]. By means of the conversion between FWHM and σ, the HSL echo signal can be represented as the following equation:

$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\sigma \tag{2}$$

$$f_\lambda(t) = \sum_{i=1}^{n_\lambda} \frac{2\sqrt{\pi \ln 2}\cdot a_{i,\lambda}}{\pi F_{i,\lambda}} exp \frac{-\left(t-\mu_{i,\lambda}\right)^2}{F_{i,\lambda}^2/(4\ln 2)} \tag{3}$$

where n_λ represents the number of Gaussian components at wavelength λ, which, in principle, should be equal across all spectral bands, and $F_{i,\lambda}$ denotes the FWHM of the ith Gaussian component at wavelength λ.

In practical measurements, the HSL system initiates digital sampling of echo pulses upon detecting the emission pulse. Given that targets are typically located at a certain distance from the laser source, a blank signal is recorded between the emission and echo pulses, which can be used to estimate the signal baseline and background noise level. Moreover, interchannel time-delay effects can cause asynchronous echo signals from the same target across different channels [53], [54], [55]. Therefore, we introduce a time offset for each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

channel, represented as the following equation:

$$f_\lambda(t) = \sum_{i=1}^{n\lambda} \frac{2\sqrt{\pi \ln 2}\cdot a_{i,\lambda}}{\pi F_{i,\lambda}} \exp \frac{-\left(t-\Delta t_{i,\lambda}-\mu_{i,\lambda}\right)^2}{F_{i,\lambda}^2/(4l\ln 2)} + B_{i\lambda} + N_{i\lambda} \tag{4}$$

$$B_{i,\lambda} = \frac{1}{200}\sum_{k=1}^{200} A_{k,\lambda} \tag{5}$$

where $\Delta t_{i,\lambda}$, $B_{i,\lambda}$, and $N_{i,\lambda}$ represent the time offset, signal baseline, and background noise of the ith Gaussian component at wavelength λ, respectively, and $A_{k,\lambda}$ represents the echo signal intensity at the kth sampling point of wavelength λ.

For simplicity, we use the first 200 sampling points of each wavelength to assess the signal baseline, as shown in (5). This enables straightforward baseline correction of the signal. Subsequently, we apply a Gaussian filter to smooth the baseline-corrected signal in order to attenuate the Gaussian white noise introduced during the signal acquisition process.

B. Waveform Alignment and Fusion via DDTW

Some targets exhibit low reflectance at specific wavelengths while showing high reflectance at others, potentially leading to signal distortion or missing information when relying on a single-wavelength channel. Given the HSL device's broad wavelength coverage, multichannel signals can compensate for each other's limitations, enabling a more comprehensive and precise characterization of targets along the laser path by leveraging the unique spectral responses across different wavelength bands. The fundamental prerequisite for realizing this approach is ensuring strict waveform alignment across all channels.

1) DDTW Algorithm: DTW is an algorithm designed to locally stretch or compress two given time series to achieve optimal alignment, finding widespread applications in speech recognition, data mining, and related fields [62], [71], [74], [80], [81]. Since the DTW algorithm focuses on the similarity of function y-values during computation and HSL data exhibit significant echo intensity variations across different spectral bands, we introduce DDTW. By calculating the first-order derivative of the signals, DDTW uses the signal's shape and variation trend as the primary feature, thereby enabling optimal signal alignment [82].

Consider two time series, a test curve P and a reference curve Q, containing n and m elements, respectively, as shown in (6). The first-order derivative at each element position is estimated using (7), resulting in (8); the first and last positions are lost in the derivation

$$P = (p_1, p_2, \ldots, p_n), Q = (q_1, q_2, \ldots, q_m) \tag{6}$$

$$x_i' = \frac{(x_i - x_{i-1}) + \left[\frac{(x_{i+1}-x_{i-1})}{2}\right]}{2} \tag{7}$$

$$P' = \left(p_2', p_3', \ldots, p_{n-1}'\right), Q' = \left(q_2', q_3', \ldots, q_{m-1}'\right) \tag{8}$$

and then, we get the distance matrix of the first derivative

$$D_{(m-2)\times(n-2)} = \begin{bmatrix} d_{1,1} & \cdots & d_{1,n-2} \\ \vdots & \ddots & \vdots \\ d_{m-2,1} & \cdots & d_{m-2,n-2} \end{bmatrix} \tag{9}$$

where $d_{i,j} = \sqrt{(p_{i+1}' - q_{j+1}')^2}$.

The purpose of DDTW is to find the optimal path W from $d_{1,1}$ to $d_{m-2,n-2}$ under certain conditions

$$W = w_1, w_2, \ldots, w_k, \ldots, w_K \tag{10}$$

$$\begin{cases} w_1 = (1,1), w_K = (m-2, n-2) \\ \mathrm{dis}\left(w_k, w_{k+1}\right) \le \sqrt{2} \\ \overrightarrow{\left(w_k, w_{k+1}\right)} \cdot \overrightarrow{(1,0)} > 0, \overrightarrow{\left(w_k, w_{k+1}\right)} \cdot \overrightarrow{(0,1)} > 0 \end{cases} \tag{11}$$

where $w_k = (i, j)_k$. K is satisfied that

$$\max\{(m-2), (n-2)\} \le K \le m + n - 5. \tag{12}$$

Define $\tau(i, j)$ to represent the minimum cumulative distance from (1,1) to (i, j). The optimal matching path between two time series of unequal lengths can be obtained by solving the following recursive formula using dynamic programming:

$$\tau(i,j) = d_{i,j} + \min\{\tau(i-1, j-1), \tau(i-1, j), \tau(i, j-1)\}. \tag{13}$$

Specifically, after obtaining the waveform derivatives, we used the dtw-python 1.5.1 package to calculate the alignment path, with the step pattern set to rabinerJuangStepPattern (6, "c") [80].

2) HSL Waveform Alignment and Fusion: In processing HSL signals, the initial step involves extracting an appropriate signal segment from the echo waveform. Typically, the starting point is set at the peak of the transmitted pulse, with the termination point selected based on the location of valid signal components. The selection of the termination point is critical: if positioned too far, excessive invalid waveforms will be introduced, while if positioned too close, the complete echo of the detected target may be incompletely captured.

Subsequently, the channel with the highest signal-to-noise ratio (SNR) is selected as the reference channel. In this experiment, the 784-nm channel was chosen as the reference. Each channel's echo signal is then aligned with the reference signal using the DDTW algorithm. Finally, all aligned signal waveforms are directly summed to generate a synthesized panchromatic band waveform representing all targets along the laser path.

C. Panchromatic Band AGD

1) Number and Location of Gaussian Components: For the synthesized panchromatic band, we can adopt the conventional Gaussian mixture model decomposition method, and the AGD method is adopted in this study [83]. AGD utilizes derivative spectroscopy and machine learning methodologies to provide optimized estimates of the number of Gaussian mixture components, along with their positions, FWHM, and amplitudes. Initially, we estimated the preliminary central positions of Gaussian components based on the criteria in the following equation:

$$\begin{cases} f > 3 * \mathrm{RMS}(\mathrm{noise}) \\ \dfrac{d^2 f}{dt^2} < 0 \\ \dfrac{d^3 f}{dt^3} = 0 \\ \dfrac{d^4 f}{dt^4} > 0 \end{cases} \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

where we set the signal detection threshold as three times the root mean square (rms) of the background noise, and signals below this threshold are not detected. The second derivative of the signal being less than 0 ensures that the search location is within the $(\mu_i-\sigma_i,\mu_i+\sigma_i)$ interval. Further constraints on the third and fourth derivatives enable more precise identification of peak locations.

2) Widths of Gaussian Components: Once the number and positions of the Gaussian components are determined, we can estimate the width of each component. For an individual Gaussian component (hypothesis $a = 1$), its second derivative is given by (15). Substituting the component's center position $t = \mu$ into (15) yields (16), where the second derivative reaches its minimum value. The FWHM of the component can then be calculated through a transformation, as shown in the following equation:

$$\frac{d^2f}{dt^2} = \left[\frac{64\ln^2 2\cdot(t-\mu)^2}{F^4} - \frac{8\ln 2}{F^2}\right]\cdot f(t) \tag{15}$$

$$\frac{d^2f}{dt^2}|_{t=\mu} = -\frac{16\ln 2\cdot\sqrt{\pi ln2}}{\pi F^3} \tag{16}$$

$$F = \sqrt[3]{-\frac{16\ln 2\cdot\sqrt{\pi\ln 2}}{\pi\cdot\frac{d^2f}{dt^2}|_{t=\mu}}} \tag{17}$$

where μ represents the central position derived from step 1) of Section III-C and the value of $(d^2f/dt^2)|_{t=\mu}$ is calculated from the waveform.

3) Amplitudes of Gaussian Components: In the echo waveform of a laser beam, Gaussian components returned from different targets are often not completely independent, but rather superimposed, which means that we cannot simply assess component amplitudes through the original echo intensity. In fact, the echo intensity at each position is a linear superposition of multiple Gaussian components, which can be represented by the following equation:

$$\begin{bmatrix} G_{11} & \cdots & G_{1n} \\ \vdots & \ddots & \vdots \\ G_{n1} & \cdots & G_{nn} \end{bmatrix}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} A_1 \\ \vdots \\ A_n \end{bmatrix} \tag{18a}$$

where matrices a and $\mathbf{A}$ represent the original and mixed amplitudes of each component, respectively. $\mathbf{G}$ is an $n\times n$ matrix, and each element of $\mathbf{G}$ is defined as

$$G_{ij} = \frac{1}{\sqrt{2\pi}\sigma_j}\exp\frac{-\left(\mu_i-\mu_j\right)^2}{2\sigma_j^2}. \tag{18b}$$

Solving (18a) is equivalent to minimizing the objective function

$$\|Ga-A\|^2 = (Ga-A)^T(Ga-A). \tag{19}$$

Taking the derivative of a and setting it to zero gives the normal equation

$$G^T(Ga-A) = 0. \tag{20}$$

Then, we get the best estimate of original amplitudes

$$a = \left(G^TG\right)^{-1}G^TA. \tag{21}$$

Finally, recheck the decomposition results and remove components that do not satisfy $a > 3^*$rms (noise). Also, ensure that the overall fit waveform has rms error (RMSE) $< 3^*$std (noise). Otherwise, adjust the regularization parameter logα and perform the decomposition again.

D. All Channels Parameter Estimation via Inverse DDTW

Using the Gaussian mixture model in (3), we perform Levenberg–Marquardt (LM) iterative optimization for each channel to find the optimal fitting parameters [84]. LM optimization is a nonlinear least-squares method, and its results are highly dependent on the initial parameter selection. If the initial parameters are inappropriate, the optimization process may get trapped in a local minimum and fail to find the global optimum [85], [86].

The decomposition results of the synthesized panchromatic band waveform can effectively characterize all targets along the laser path. The warping paths then utilized to map the Gaussian component parameters identified in the panchromatic band back to the time axis of each original channel. Specifically, for the center positions of each Gaussian component in the panchromatic band, the corresponding time points in the original channels are found by searching the DDTW warping path, thereby obtaining the initial center positions of this Gaussian component in each channel. The intensity corresponding to the original central position is regarded as the initial amplitude. The FWHM is calculated as the time interval between the mapped left and right half-maximum positions in each channel. Through the inverse transformation of DDTW, we can precisely recover the position of each component in the panchromatic band for each channel, thereby providing unique and reliable initial parameters for the LM optimization of each channel. The optimization results should satisfy the following criteria.

1) The number of Gaussian components should be the same as in the synthesized panchromatic band.
2) The center position of each component should not deviate from the initial value by more than 50% of the FWHM.
3) RMSE should be less than three times the standard deviation of the noise.

E. Method Comparison

To demonstrate the superior performance of the proposed DWF method in time-delay effect correction and multiecho signal decomposition for HSL systems, we selected the MIWD method proposed by Wang et al. [55] for comparison. Equal-weight accumulation was selected as the waveform accumulation method for MIWD, with manual time-delay alignment performed for each channel. Other steps were strictly followed according to the workflow. In contrast, the data-driven DWF method proposed in this article is based on analyzing the intrinsic features of the data itself for time-delay correction. Additionally, a single-channel GD is performed for each channel as a reference.

F. Accuracy Assessment

Four metrics were used to evaluate the performance of the waveform decomposition algorithm. The coefficient of determination R^2 and RMSE were used to assess the decomposition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

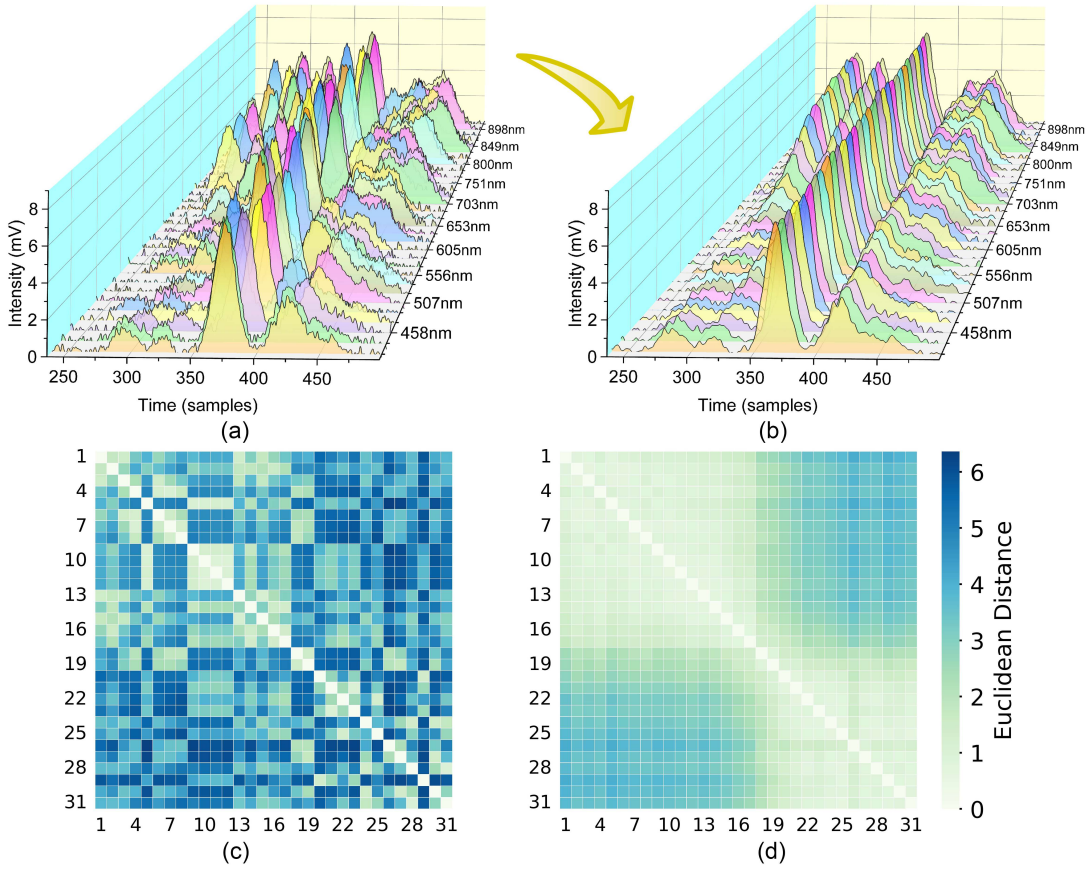

Fig. 4. Comparison of Group II data before and after correction for interchannel time-delay effects. (a) and (b) Waveform curves of all channels before and after correction. (c) and (d) Euclidean distance of the normalized curve between each channel before and after correction.

results, while the relative neighbor distance error (RNDE) [49] was used to evaluate the capability of extracting spatial information in the synthesized panchromatic band

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{22}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{23}$$

$$\text{RNDE} = \frac{|D_{\text{true}} - D_{\text{measured}}|}{D_{\text{true}}} \times 100\% \tag{24}$$

where y_i is the measured raw value; $\hat{y}_i$ stands for the estimated value of fit waveform; $\bar{y}$ denotes the average value of measured data; and D_{true} and D_{measured} represent the true distance between two targets and the measured distance obtained through waveform decomposition, respectively.

To verify the difference between the spectra obtained from waveform decomposition and the true target spectra, a multiband LiDAR spectral similarity index (MLSI) specific to HSL systems was proposed, defined as

$$\text{MLSI} = \frac{1}{m} \sum_{i=1}^{m} \frac{\min(r_{1,i}, r_{2,i})}{\max(r_{1,i}, r_{2,i})} \tag{25a}$$

where MLSI measures the similarity between spectra 1 and 2; m represents the number of spectral channels; and $r_{1,i}$ and $r_{2,i}$ represent the reflectance of spectra 1 and 2 in the ith channel, respectively. The value of MLSI ranges from [0, 1], with higher values representing closer spectra. For the measured data waveform, r_i is defined as

$$r_i = \frac{\int_{\mu_i - 3\sigma_i}^{\mu_i + 3\sigma_i} f_{\text{target}}(t)}{\int_{\mu_i - 3\sigma_i}^{\mu_i + 3\sigma_i} f_{\text{SRP}}(t)} \tag{25b}$$

where μ and σ are the center position and standard deviation of LiDAR signal components, respectively, and $f_{\text{target}}(t)$ and $f_{\text{SRP}}(t)$ are the echo signal waveforms of the target and the 99% SRP at the same distance, respectively.

IV. Results

A. Correction Results of Interchannel Time-Delay Effect

We created three sets of signal waveforms with different levels of interchannel time delays in the simulated dataset. Specifically, random time delays of ±10, ±20, and ±30 samples were applied to each channel in Data Groups I–III, respectively. Data Group II is used as an example in the main text, and more detailed pictures can be found in the supplementary material (Figs. S1 and S2). Fig. 4 illustrates the comparison of Group II data before and after correction. Fig. 4(a) displays the raw data waveforms of 31 channels, with the horizontal axis representing time in data acquisition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AVERAGE EUCLIDEAN DISTANCE OF THE NORMALIZED CURVES BETWEEN CHANNELS BEFORE AND AFTER CORRECTION FOR DATA GROUPS I–III

Data Group number	Random inter-channel time delay (samples)	Average Euclidean distance (mV)	
		before correction	after correction
I	± 10	2.88	1.96
II	± 20	3.96	2.01
III	± 30	4.22	2.07

units, where the sampling interval between two adjacent points is 0.2 ns. We can observe that the signal peak positions across different channels are notably misaligned, and certain components exhibit very low signal intensity in specific spectral bands, introducing significant uncertainty in information extraction. Fig. 4(b) shows the waveforms after alignment using the DDTW step. It is evident that the component centers of each channel are nearly identical. After normalizing the signal intensity, the Euclidean distance between the waveforms of different channels is calculated. Fig. 4(c) and (d) displays the results before and after correction, respectively, showing that the signal alignment significantly reduces the waveform differences between channels. In fact, spectrally adjacent channels tend to have waveform curves with higher correlations, and the low-value regions in the upper left and lower right corners of the corrected data in Fig. 4(d) clearly reflect this characteristic.

To verify the robustness of the DDTW alignment algorithm, we applied it to three groups of simulated data and statistically analyzed the results. Table II presents the specific impact of signal alignment on the average Euclidean distance between waveforms across channels in the three datasets. Prior to correction, as the time-delay effects intensified (random time delays: ±10, ±20, and ±30), the average interchannel Euclidean distances increased significantly (2.88, 3.96, and 4.22). However, after correction, it returned to a stable level (1.96, 2.01, and 2.07), and this stable value depends on the inherent spectral intensity differences between channels. These results demonstrate that the DDTW method can achieve temporal alignment of HSL signal waveforms across different channels and is adaptable to various conditions, thus laying a solid foundation for subsequent signal decomposition and information extraction.

The waveform alignment strategy of the DDTW algorithm is not a simple time-axis translation but involves feature point detection and local scaling of signals for each channel based on waveform shape. The ability to correctly match feature points across all channels is crucial to successful waveform alignment. Fig. 5 visualizes the alignment paths between signals of each channel and the reference channel. In each subplot, the orange curve above represents the first-order derivative of the current channel's waveform, while the magenta curve below shows the first-order derivative of the reference channel's waveform. The dashed line in the middle indicates the obtained alignment path. As apparent in Fig. 5, even in channels with obviously weak individual component responses and time-delay effects, the DDTW algorithm can still successfully match feature points in the HSL signal. By employing first-order derivatives as the alignment data source, we can more effectively capture waveform features such as peaks, valleys, inflection points, and subtle shape characteristics, leading to a more refined correction.

B. HSL Waveform Decomposition and Information Extraction Results

After aligning and accumulating the waveforms of each channel, the synthesized panchromatic band was obtained. Fig. 6 shows the waveform curves of the panchromatic band for the three datasets, along with the Gaussian components obtained from AGD single-band waveform decomposition. As all signals from the 31 channels are fused, the intensity of each component significantly increases. It is evident that the five components in each dataset exhibit clear peak shapes, all of which are successfully detected by the waveform decomposition algorithm. Table III provides a detailed list of the decomposition results for the three data groups, where the central position μ and full width at half maximum F of each Gaussian component are important initial parameters for LM optimization. By using the alignment paths shown in Fig. 5, the Gaussian component parameters of the panchromatic band are reverse-transformed back to each channel, allowing us to obtain the initial theoretical values of the five Gaussian components across the 31 channels.

In terms of spatial geometry information extraction, we evaluated the ranging accuracy between any two of the five targets using the synthesized panchromatic band, with the percentage-based RNDE as the evaluation metric. The results are shown in Table IV. The average RNDE for the three datasets is 1.33%, 1.24%, and 2.51%, with the maximum ranging error between any two targets being 5.43%.

Waveform decomposition of all channels can be performed by using the initial parameters of each channel obtained by inverse DDTW. Fig. 7 presents the decomposition results obtained from LM optimization applied to all channels, using R^2 and RMSE as metrics to evaluate the decomposition accuracy. Different Gaussian components are represented by different colors, with the black solid line representing the sum of all Gaussian components and the blue scattered points representing the original waveform data points. Since the initial parameters were based on the panchromatic band, the number and position of the Gaussian components for all channels were accurately determined. Even components with low intensity values in the range from 425 to 670 nm were successfully detected. For comparison, the single-channel GD produced one or two missed detections in 17 channels (as shown in Fig. S3). Moreover, despite the increasing time-delay effects between channels in the three datasets, the accuracy of DWF did not show a significant reduction (average R^2 = 0.984, 0.983, and 0.984; average RMSE = 0.211, 0.214, and 0.210, as shown in Fig. S2). These results demonstrate that the DWF algorithm possesses strong robustness.

The information obtained from HSL echoes primarily consists of three parts: spatial ranging information can be derived from the center positions of signal components, spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

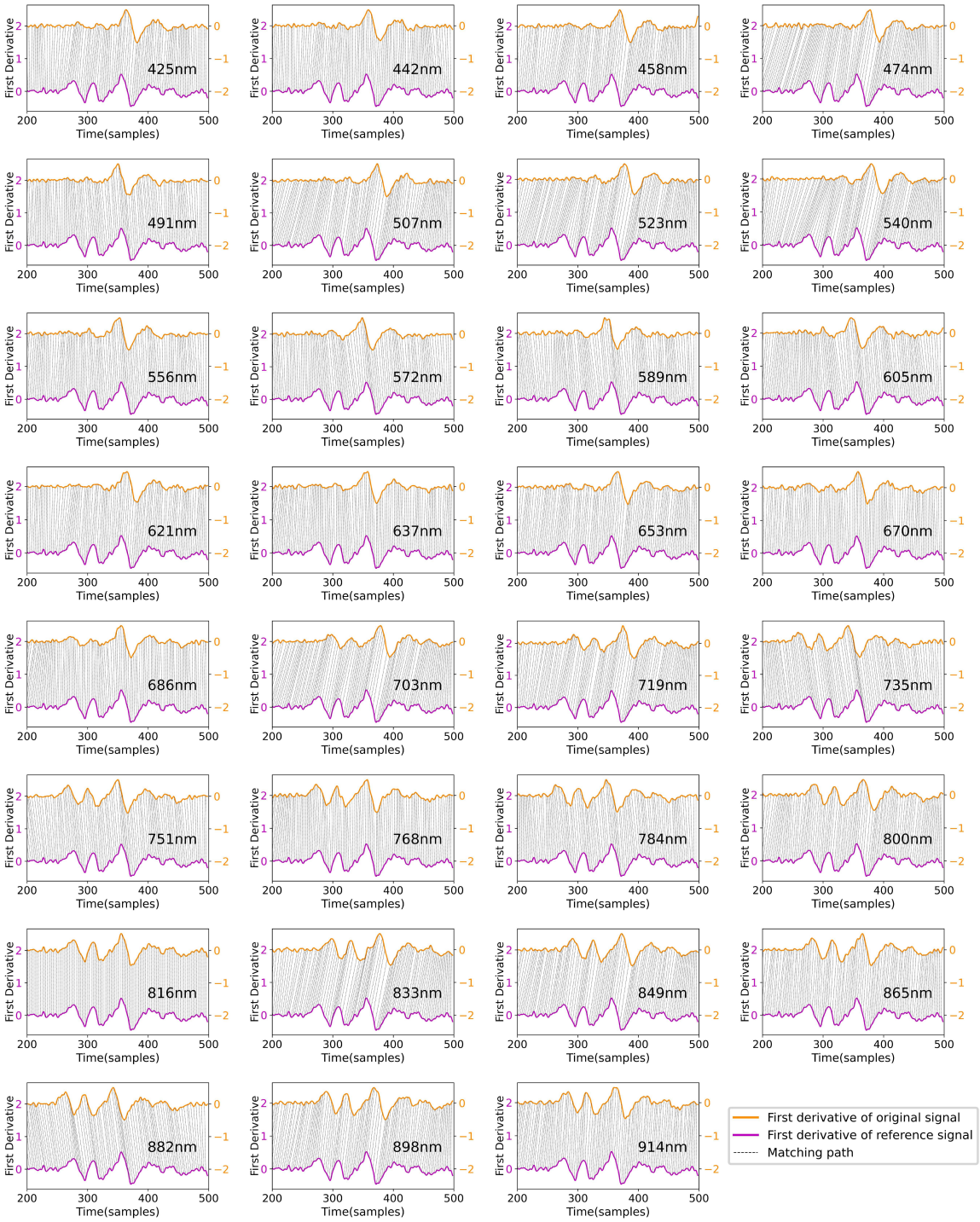

Fig. 5. Alignment path of the first derivative waveform of each channel.

information can be extracted from the echo intensities of different channels, and physical characteristics such as surface roughness can be characterized to some extent through signal parameters like FWHM. Table S1 presents the Gaussian component parameters for each channel in the decomposition results of Group II.

Since we use the central position of synthesized panchromatic band for spatial geometry extraction, we primarily focus on the peak amplitude (A) and FWHM (F) in each channel. These two parameters are closely related to the physical properties of the targets, such as spectral reflectance and surface roughness. The true value of A depends on the original

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

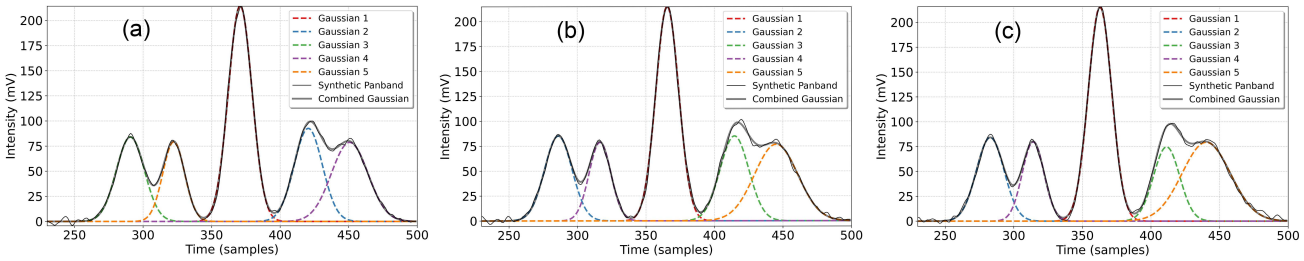

Fig. 6. Synthetic panchromatic band waveforms and decomposition results of (a) Data Group I, (b) Data Group II, and (c) Data Group III.

TABLE III
SPECIFIC PARAMETERS OF THE DECOMPOSITION RESULTS IN FIG. 6

Gaussian component	Group I			Group II			Group III		
	A	F	μ	A	F	μ	A	F	μ
1	84.09	24.04	290.73	85.27	23.58	286.06	84.28	23.66	282.86
2	79.25	18.77	322.36	78.82	18.74	316.46	79.77	18.74	314.13
3	214.82	20.34	370.85	215.23	19.71	365.64	215.49	20.22	362.86
4	92.63	24.23	420.53	85.16	24.67	414.62	74.78	22.45	411.38
5	78.69	31.43	450.70	76.78	37.09	445.06	79.34	41.38	440.27

TABLE IV
RESULTS OF RANGING PERFORMANCE EVALUATION BETWEEN ANY TWO TARGETS

Target pair	RNDE (%)		
	Group I	Group II	Group III
1-2	5.43	1.33	4.23
1-3	0.15	0.53	0.00
1-4	0.15	1.11	1.14
1-5	0.02	0.63	1.62
2-3	3.02	1.64	2.54
2-4	1.83	1.84	2.75
2-5	1.28	1.08	2.97
3-4	0.64	2.04	2.96
3-5	0.19	0.72	3.24
4-5	0.57	1.47	3.70
average	1.33	1.24	2.51

spectral curves of the five targets shown in Fig. 8(a), and the true value of FWHM can be found in Table I.

In terms of spectral information extraction, many previous studies have used the peak amplitude of the Gaussian components in the echo signal as the reflection intensity at that position [16], [28], [51], [52]. However, the total laser echo energy contained in the echo signal should be determined by both the height and width of the waveform. For each Gaussian component, we can replace the peak intensity with the integrated area of the waveform within the range $\mu \pm 3\sigma$, which can better represent the spectral signature of the targets [60], as shown in (25b). Fig. 8(b) and (c) shows the extracted spectral curves for the four targets (excluding SRP) from the 31-channel waveforms in Fig. 7, using both the peak intensity and integrated area methods. Comparing the spectral curves, the peak intensity spectra in Fig. 8(b) are more stable in the low-reflectance region around 440 nm, whereas in the high-reflectance region from 700 to 900 nm, the integrated area spectra in Fig. 8(c) closely resemble the original spectra shown in Fig. 8(a). In the near infrared region of the spectrum, the reflectance of plywood should be higher than that of leaf and lower than that of petal. The integrated area spectra preserve this feature well, while the peak intensity spectra show errors in this region. Fig. 8(d) shows the MLSI evaluation results of the similarity between the spectra extracted by two methods and the original spectra. Compared to peak intensity spectra, the MLSI of integrated area spectra improved from 0.827 to 0.946 relative to the true spectra, while the Euclidean distance decreased from 0.553 to 0.174. The results indicate that using the integrated area to represent the total energy of the echo signal yields better results in spectral curve extraction.

Regarding FWHM, for most channels, the decomposition results are highly consistent with the true values. Fig. 9 visualizes the decomposition results of the five Gaussian components across the 31 channels, with the blue dashed lines representing the true values. The RMSEs for the fits are 1.96, 1.32, 0.18, 0.98, and 2.37, indicating that the waveform decomposition results exhibit high stability. The data deviations are primarily concentrated in the wavelength range from 400 to 600 nm, with the FWHM decomposition results stabilizing near the true values for wavelengths greater than 600 nm. This is likely due to the LM algorithm's lack of sensitivity to small Gaussian components with low reflectance and closely spaced central positions, as the overall fit accuracy mainly depends on the high-reflectance regions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

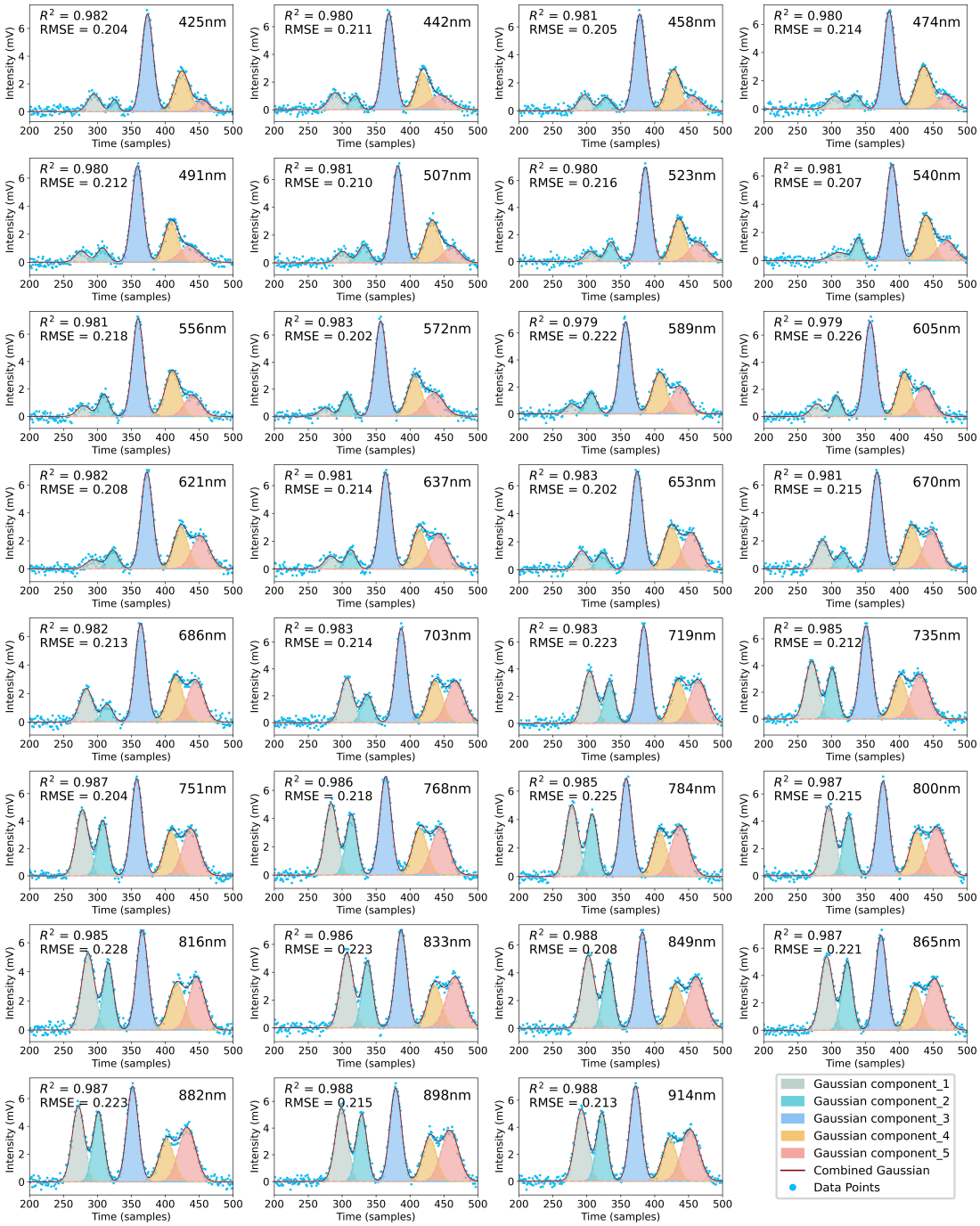

Fig. 7. Final decomposition results of the DWF method for Data Group II.

C. Model Comparison Results

The DWF method proposed in this study was applied to waveform decomposition of measured data and compared with single-channel GD and MIWD methods. The Gaussian components and accuracy evaluation metric are shown in Fig. S4. Due to the obvious noise in the measured data, the single-channel GD method showed oversegmentation or missed detection in 12 channels [Fig. S4(a)]. Benefiting from the restriction on the number of Gaussian components, the MIWD method detected three components in each channel, but some deviations occurred at 605, 637, and 653 nm. These errors may be attributed to the lack of accurate initial estimation and insufficient constraints for component center positions and FWHM in each channel. The time delays between channels in the HSL system are influenced by multiple components, such as fibers, gratings, and photoelectric converters, making it difficult to theoretically calculate the time delays precisely. When a channel is not perfectly aligned or the signal waveform is suboptimal, it is more prone to getting stuck in a local optimum.

In contrast, the DWF method proposed in this study extracts the shape features of each channel's signal using the first derivative of the waveform and uses these features as the basis for time-delay correction. Additionally, the decomposition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

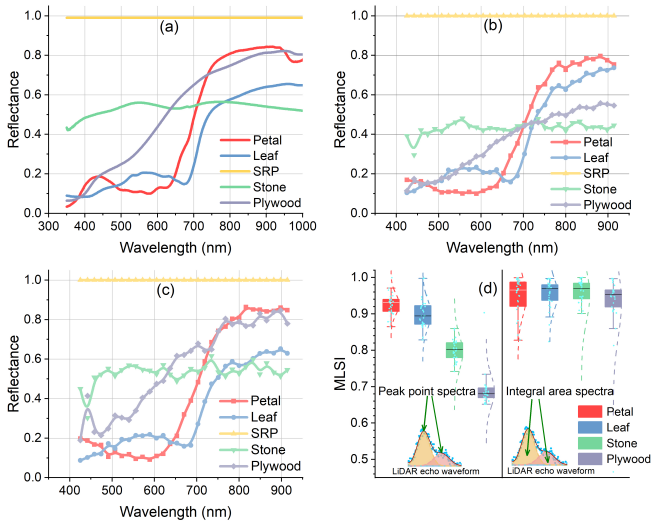

Fig. 8. Spectral curve extraction and similarity comparison results. (a) Original spectral curves. (b) Spectral curves extracted by peak points method. (c) Spectral curves extracted by integral area method. (d) Boxplot comparing the similarity between extracted and original spectral curves.

Fig. 9. Statistical stacking plot of FWHM decomposition results for each channel.

results of the synthesized panchromatic band are used with an alignment path reverse transformation strategy to provide unique initial parameters for each channel, ensuring that the center position of each Gaussian component in the final LM optimization result does not deviate from the initial parameters by more than 0.5 ∗ FWHM. These measures allow the DWF method to produce consistent signal component quantities and accurate center positions across all channels. Fig. S4(c) shows the performance of the DWF method on the measured data, with $R^2 > 0.9$ in 29 out of the 31 channels and the maximum RMSE not exceeding 0.6. Table V presents a comparison of the three methods in ranging accuracy of measured dataset, showing that the DWF method is always optimal in relative distance measurements between any two targets. The ranging results obtained by the single-channel GD method in different channels fluctuate significantly. Compared with MIWD in all

TABLE V
COMPARISON OF RANGING ACCURACY AMONG GD, MIWD, AND DWF METHODS ON THE MEASURED DATASET

Method	Target pair	Measured distance (cm)	RNDE (%)	AVERAGE RNDE (%)
GD	1-2	49.0~65.1	1.13~15.54	5.15
	1-3	148.2~168.9	0.02~10.38	
	2-3	86.4~119.9	0.29~26.21	
MIWD	1-2	55.6	4.14	5.11
	1-3	158.1	3.33	
	2-3	102.4	7.87	
DWF	1-2	56.5	2.59	2.27
	1-3	149.6	2.22	
	2-3	93.1	2.00	

measured channels, the DWF method improved R^2 from 0.94 to 0.97, reduced RMSE from 0.31 to 0.21, and decreased RNDE from 5.11% to 2.27%.

V. DISCUSSION

Some previous studies have utilized multichannel waveform information for complementation and enhancement when performing HSL signal decomposition, aiming to improve the recognition rate of weak components [49], [53], [54], [55]. Since the signal components in different channels are not perfectly synchronized in time, these studies generally adopt methods such as sorting signal components and taking median values to provide the most probable initial parameters for LM optimization [49], [50], [54]. In this study, we propose a DWF method based on waveform shape features, which fills the gap in time series correction and parameter extraction of HSL signals.

One advantage of the DWF method is its ability to correct interchannel time-delay effects of HSL signals, as described in Section III-B. By aligning time series features through the first-order derivatives of waveforms across different channels, the DWF method enables better identification of the same signal components across channels. Supercontinuum laser originates from nonlinear spectral broadening of narrow-wavelength laser [38], [39], [40]. Specifically, the HSL-II instrument uses a source with an original wavelength of 1064 nm. Consequently, the spectral energy density of the supercontinuum laser peaks at 1064 nm and oscillates with decreasing amplitude as wavelength increases or decreases, eventually vanishing beyond 400 and 2400 nm. Additionally, the cathode radiant sensitivity and quantum efficiency of PMT also vary with wavelength, as shown in Fig. 10. These hardware factors, combined with the target's inherent spectral characteristics, collectively lead to inconsistent signal responses across different wavelengths. Therefore, the missed detection of individual signals may occur in a single channel. After aligning all channel signals, high-response channels can compensate for low-response channels, resulting in more robust signal component positioning. The transition from Fig. 4(a) and (b) illustrates that previously disordered waveforms become well-aligned after applying the DWF

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

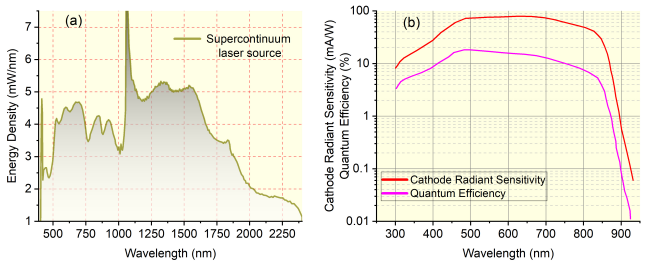

Fig. 10. (a) Energy density distribution of the supercontinuum laser source. (b) Cathode radiant sensitivity and quantum efficiency of the PMT element.

algorithm's alignment step. The weak signal groups appearing in channels below 600 nm are effectively processed. Results in Table II confirm that the alignment strategy remains effective across different time-delay levels. This is of great significance for further signal processing and information extraction.

The inverse DDTW method was applied to generate high-quality initial parameters for each channel, as described in Section III-D. Theoretically, for a specific laser path, the number and central positions of echo components across different channels should be identical. However, single-channel waveforms, due to their narrow spectral range, are difficult to detect low-reflectivity targets at specific wavelengths. Fortunately, the excellent performance of the signal alignment strategy provides a foundation for synthesizing panchromatic band LiDAR waveforms. The panchromatic band waveform essentially superimposes echo signals of each target across all spectral bands, utilizing high-reflectivity channels to supplement low-reflectivity channels, thereby significantly reducing the likelihood of missed detection. Both MIWD and DWF methods use signals of multiple channels to accumulate [55], but different time-delay alignment methods lead to different initial parameters of each channel input to the LM model. When modeling HSL waveforms as a Gaussian mixture model composed of n Gaussian components, we are essentially addressing a nonlinear optimization problem with 3n parameters. Due to the nonorthogonal nature of Gaussian functions, the parameter space is nonconvex, making it impossible to obtain a unique correct result and leaving us with only a series of local optimal solutions [83]. Therefore, the final decomposition results are closely related to the initial parameter selection [49], [54], [55], [77]. By inputting the number of Gaussian components, their center positions, and FWHM obtained through inverse DDTW into the LM algorithm, we can maximize the consistency of decomposition results across channels and avoid falling into local optima. Furthermore, some scholars have proposed methods such as nonparametric hybrid models and end-to-end neural networks, which can achieve waveform decomposition without assuming waveform distribution and initial parameters [76], [87].

One limitation of this method is that the final results of each channel are sensitively dependent on the decomposition results of the synthesized panchromatic band. Since both regularization parameters and noise thresholds are set empirically, misjudgments may occur when detecting waveforms of different shapes from various target types. The sensitivity analysis results (Fig. S5) indicate that when the sigma value of Gaussian filter is below 0.5, different regularization parameters (alpha) may introduce variability in the outcomes. As sigma increases, the average R^2 of waveform fitting gradually improves, and R^2 tends to stabilize when sigma exceeds 2.5. By selecting an appropriate filtering intensity based on the characteristics of the raw data, better decomposition results can be achieved while avoiding both noise interference and excessive smoothing. Another limitation is computational complexity. For traditional GD methods, when decomposing a single-channel signal containing k Gaussian peaks, the computational complexity is $O(N \cdot k^2 \cdot T)$, where N is the number of data points and T is the number of iterations in the optimization algorithm. In practical applications, T is typically controlled by convergence criteria and generally ranges from tens to hundreds of iterations. For multichannel signals with C channels, traditional methods require C independent decompositions, resulting in a total computational complexity of $O(C \cdot N \cdot k^2 \cdot T)$. In contrast, the DWF method requires DTW alignment [complexity $O(C \cdot N^2)$] followed by a single GD of the superimposed signal [complexity $O(N \cdot k^2 \cdot T)$]. The computational efficiency of the DWF method can be significantly improved by reducing N through signal downsampling or by employing FastDTW algorithms to reduce DTW complexity to near-linear levels. Furthermore, for complex signals with numerous peaks, the DWF method may provide additional computational efficiency advantages by reducing the required number of GDs.

In particular, different laser footprint sizes may lead to scale effects in echo shapes, a concern that warrants attention in future large-scale remote sensing applications. Moreover, all channel waveforms are aligned with the reference channel by the DDTW step, but the reference channel itself may still have a certain time delay. When the number of channels increases or the coverage range becomes wider, the waveform characteristics of infrared channels may differ from those of visible channels. How to choose the best reference channel or design a better optical path structure is worthy of further study.

Finally, a more fundamental approach to solving the signal quality issues is to develop more advanced HSL systems. Existing instruments are mostly complex prototypes with components that are not sufficiently miniaturized, making it difficult to achieve wide-area rapid detection. Currently, there are well-established single-band LiDAR systems that have demonstrated superior performance in terms of detection range and scanning speed. Therefore, future research on HSL instrument miniaturization and airborne platform applications is promising, as it will provide a new and practical tool for Earth observation.

VI. Conclusion

In the data processing of existing HSL systems, effective correction of interchannel time-delay effects is necessary. However, most existing studies have focused on how to arrange and select data to obtain valid information, rather than how to correct the data itself. Therefore, this article proposes a DWF method based on the shape features of each channel's

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

waveform for HSL signal alignment and decomposition and demonstrates its effectiveness using both simulated and measured data. The main conclusions are as follows.

1) The DWF algorithm effectively aligns the signals, and its performance remains stable despite increasing time-delay effects across channels.

2) We introduced a new spectral similarity index MLSI for HSL echo signals, demonstrating that spectra extracted from the integrated area of the waveform are closer to the original spectra than those extracted from the peak points.

3) The DWF method provides unique initial parameters in each channel, demonstrating better performance in signal component parameter extraction and spatial ranging.

This method offers a new approach for multiband LiDAR data processing and may also inspire time series signal analysis in other fields. Moving forward, we will soon develop HSL equipment suitable for airborne platforms, featuring more powerful supercontinuum lasers and larger telescope apertures, which can be used for large-scale Earth observation missions.

REFERENCES

[1] Q. Guo et al., "LiDAR boosts 3D ecological observations and modelings: A review and perspective," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 232–257, Mar. 2021, doi: 10.1109/MGRS.2020.3032713.

[2] M. A. Hofton, E. Malavassi, and J. B. Blair, "Quantifying recent pyroclastic and lava flows at arenal volcano, Costa rica, using medium-footprint LiDAR," *Geophys. Res. Lett.*, vol. 33, no. 21, Nov. 2006, Art. no. L21306, doi: 10.1029/2006gl027822.

[3] M. A. Hofton, J. B. Minster, and J. B. Blair, "Decomposition of laser altimeter waveforms," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 4, pp. 1989–1996, Jul. 2000, doi: 10.1109/36.851780.

[4] C. Mallet and F. Bretar, "Full-waveform topographic LiDAR: State-of-the-art," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 1, pp. 1–16, Jan. 2009, doi: 10.1016/j.isprsjprs.2008.09.007.

[5] N. C. Coops et al., "Estimating canopy structure of douglas-fir forest stands from discrete-return LiDAR," *Trees*, vol. 21, no. 3, pp. 295–310, May 2007, doi: 10.1007/s00468-006-0119-6.

[6] S. Luo et al., "Estimating forest aboveground biomass using small-footprint full-waveform airborne LiDAR data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 83, Nov. 2019, Art. no. 101922, doi: 10.1016/j.jag.2019.101922.

[7] F. Pirotti, G. Laurin, A. Vettore, A. Masiero, and R. Valentini, "Small footprint full-waveform metrics contribution to the prediction of biomass in tropical forests," *Remote Sens.*, vol. 6, no. 10, pp. 9576–9599, Oct. 2014, doi: 10.3390/rs6109576.

[8] X. Shang and P. Chazette, "Interest of a full-waveform flown UV LiDAR to derive forest vertical structures and aboveground carbon," *Forests*, vol. 5, no. 6, pp. 1454–1480, Jun. 2014, doi: 10.3390/f5061454.

[9] T. Yao et al., "Measuring forest structure and biomass in new England forest stands using echidna ground-based LiDAR," *Remote Sens. Environ.*, vol. 115, no. 11, pp. 2965–2974, Nov. 2011, doi: 10.1016/j.rse.2010.03.019.

[10] C. Alexander, B. Deák, A. Kania, W. Mücke, and H. Heilmeier, "Classification of vegetation in an open landscape using full-waveform airborne laser scanner data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 41, pp. 76–87, Sep. 2015, doi: 10.1016/j.jag.2015.04.014.

[11] L. Cao, N. C. Coops, J. L. Innes, J. Dai, H. Ruan, and G. She, "Tree species classification in subtropical forests using small-footprint full-waveform LiDAR data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 49, pp. 39–51, Jul. 2016, doi: 10.1016/j.jag.2016.01.007.

[12] S. M. Marselis et al., "Evaluating the potential of full-waveform LiDAR for mapping pan-tropical tree species richness," *Global Ecol. Biogeography*, vol. 29, no. 10, pp. 1799–1816, Oct. 2020, doi: 10.1111/geb.13158.

[13] S. M. Marselis et al., "Exploring the relation between remotely sensed vertical canopy structure and tree species diversity in Gabon," *Environ. Res. Lett.*, vol. 14, no. 9, Sep. 2019, Art. no. 094013, doi: 10.1088/1748-9326/ab2dcd.

[14] W. Yao, P. Krzystek, and M. Heurich, "Tree species classification and estimation of stem volume and DBH based on single tree extraction by exploiting airborne full-waveform LiDAR data," *Remote Sens. Environ.*, vol. 123, pp. 368–380, Aug. 2012, doi: 10.1016/j.rse.2012.03.027.

[15] K. Bi et al., "Simultaneous extraction of plant 3-D biochemical and structural parameters using hyperspectral LiDAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3025321.

[16] K. Bi, S. Xiao, S. Gao, C. Zhang, N. Huang, and Z. Niu, "Estimating vertical chlorophyll concentrations in maize in different health states using hyperspectral LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8125–8133, Nov. 2020, doi: 10.1109/TGRS.2020.2987436.

[17] X. Zhu, T. Wang, A. K. Skidmore, R. Darvishzadeh, K. O. Niemann, and J. Liu, "Canopy leaf water content estimated using terrestrial LiDAR," *Agricult. Forest Meteorol.*, vol. 232, pp. 152–162, Jan. 2017, doi: 10.1016/j.agrformet.2016.08.016.

[18] C. Alexander, K. Tansey, J. Kaduk, D. Holland, and N. J. Tate, "Backscatter coefficient as an attribute for the classification of full-waveform airborne laser scanning data in urban areas," *ISPRS J. Photogramm. Remote Sens.*, vol. 65, no. 5, pp. 423–432, Sep. 2010, doi: 10.1016/j.isprsjprs.2010.05.002.

[19] F. Bretar, A. Chauve, J.-S. Bailly, C. Mallet, and A. Jacome, "Terrain surfaces and 3-D landcover classification from small footprint full-waveform LiDAR data: Application to badlands," *Hydrol. Earth Syst. Sci.*, vol. 13, no. 8, pp. 1531–1544, Aug. 2009, doi: 10.5194/hess-13-1531-2009.

[20] C. Mallet, F. Bretar, M. Roux, U. Soergel, and C. Heipke, "Relevance assessment of full-waveform LiDAR data for urban area classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 6, pp. S71–S84, Dec. 2011, doi: 10.1016/j.isprsjprs.2011.09.008.

[21] W. Y. Yan, A. Shaker, and N. El-Ashmawy, "Urban land cover classification using airborne LiDAR data: A review," *Remote Sens. Environ.*, vol. 158, pp. 295–310, Mar. 2015, doi: 10.1016/j.rse.2014.11.001.

[22] R. Schwarz, G. Mandlburger, M. Pfennigbauer, and N. Pfeifer, "Design and evaluation of a full-wave surface and bottom-detection algorithm for LiDAR bathymetry of very shallow waters," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 1–10, Apr. 2019, doi: 10.1016/j.isprsjprs.2019.02.002.

[23] D. Wang, S. Xing, Y. He, J. Yu, Q. Xu, and P. Li, "Evaluation of a new lightweight UAV-borne topo-bathymetric LiDAR for shallow water bathymetry and object detection," *Sensors*, vol. 22, no. 4, p. 1379, Feb. 2022, doi: 10.3390/s22041379.

[24] L. Wu, Y. Chen, Y. Le, Y. Qian, D. Zhang, and L. Wang, "A high-precision fusion bathymetry of multi-channel waveform curvature for bathymetric LiDAR systems," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103770, doi: 10.1016/j.jag.2024.103770.

[25] M. Doneus, C. Briese, M. Fera, and M. Janner, "Archaeological prospection of forested areas using full-waveform airborne laser scanning," *J. Archaeological Sci.*, vol. 35, no. 4, pp. 882–893, Apr. 2008, doi: 10.1016/j.jas.2007.06.013.

[26] R. Lasaponara and N. Masini, "Full-waveform airborne laser scanning for the detection of medieval archaeological microtopographic relief," *J. Cultural Heritage*, vol. 10, pp. e78–e82, Dec. 2009, doi: 10.1016/j.culher.2009.10.004.

[27] C. Sevara, M. Wieser, M. Doneus, and N. Pfeifer, "Relative radiometric calibration of airborne LiDAR data for archaeological applications," *Remote Sens.*, vol. 11, no. 8, p. 945, Apr. 2019, doi: 10.3390/rs11080945.

[28] J. Bai et al., "An exploration, analysis, and correction of the distance effect on terrestrial hyperspectral LiDAR data," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 60–83, Apr. 2023, doi: 10.1016/j.isprsjprs.2023.03.001.

[29] W. Gong, S. L. Song, B. Zhu, S. Shi, F. Q. Li, and X. W. Cheng, "Multi-wavelength canopy LiDAR for remote sensing of vegetation: Design and system performance," *ISPRS J. Photogramm. Remote Sens.*, vol. 69, pp. 1–9, Apr. 2012, doi: 10.1016/j.isprsjprs.2012.02.001.

[30] T. Hakala, J. Suomalainen, and S. Kaasalainen, "Full waveform active hyperspectral LiDAR," in *Proc. 22nd Congr. Int.-Soc.-Photogramm.-Remote-Sens.*, Sep. 2012, pp. 459–462.

[31] P. Ray, D. Salido-Monzú, S. L. Camenzind, and A. Wieser, "Supercontinuum-based hyperspectral LiDAR for precision laser scanning," *Opt. Exp.*, vol. 31, no. 20, pp. 33486–33499, Sep. 2023, doi: 10.1364/oe.498576.

[32] H. Shao et al., "A 91-channel hyperspectral LiDAR for coal/rock classification," *ieee geosci. remote sens. lett.*, vol. 17, no. 6, pp. 1052–1056, jun. 2020, doi: 10.1109/LGRS.2019.2937720.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

[33] G. Sun et al., "32-channel hyperspectral waveform LiDAR instrument to monitor vegetation: Design and initial performance trials," *Proc. SPIE*, vol. 9263, Nov. 2014, Art. no. 926331, doi: 10.1117/12.2066788.

[34] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013, doi: 10.1109/MGRS.2013.2244672.

[35] C. Miesch, L. Poutier, V. Achard, X. Briottet, X. Lenot, and Y. Boucher, "Direct and inverse radiative transfer solutions for visible and near-infrared hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 7, pp. 1552–1562, Jul. 2005, doi: 10.1109/TGRS.2005.847793.

[36] W. Verhoef and H. Bach, "Simulation of hyperspectral and directional radiance images using coupled biophysical and atmospheric radiative transfer models," *Remote Sens. Environ.*, vol. 87, no. 1, pp. 23–41, Sep. 2003, doi: 10.1016/s0034-4257(03)00143-3.

[37] H. Ademgil and S. Haxha, "Endlessly single mode photonic crystal fiber with improved effective mode area," *Opt. Commun.*, vol. 285, no. 6, pp. 1514–1518, Mar. 2012, doi: 10.1016/j.optcom.2011.10.067.

[38] R. R. Alfano and S. L. Shapiro, "Observation of self-phase modulation and small scale filaments in crystals and glasses," *Bull. Amer. Phys. Soc.*, vol. 15, no. 4, p. 506, 1970.

[39] R. R. Alfano and S. L. Shapiro, "Emission in the region 4000 to 7000 Å via four-photon coupling in glass," *Phys. Rev. Lett.*, vol. 24, no. 11, pp. 584–587, Mar. 1970, doi: 10.1103/physrevlett.24.584.

[40] J. M. Dudley, G. Genty, and S. Coen, "Supercontinuum generation in photonic crystal fiber," *Rev. Mod. Phys.*, vol. 78, no. 4, pp. 1135–1184, Oct. 2006, doi: 10.1103/revmodphys.78.1135.

[41] L. G. Wright, W. H. Renninger, D. N. Christodoulides, and F. W. Wise, "Nonlinear multimode photonics: Nonlinear optics with many degrees of freedom," *Optica*, vol. 9, no. 7, pp. 824–841, Jul. 2022, doi: 10.1364/optica.461981.

[42] A. M. Wallace et al., "Design and evaluation of multispectral LiDAR for the recovery of arboreal parameters," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4942–4954, Aug. 2014, doi: 10.1109/TGRS.2013.2285942.

[43] Y. Chen et al., "Feasibility study of ore classification using active hyperspectral LiDAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1785–1789, Nov. 2018, doi: 10.1109/LGRS.2018.2854358.

[44] Y. Chen et al., "A 10-nm spectral resolution hyperspectral LiDAR system based on an acousto-optic tunable filter," *Sensors*, vol. 19, no. 7, p. 1620, Apr. 2019, doi: 10.3390/s19071620.

[45] H. Shao et al., "Feasibility study of forest wood-boring pest detection base on hyperspectral LiDAR," *Int. J. Remote Sens.*, vol. 45, no. 6, pp. 1926–1948, Mar. 2024, doi: 10.1080/01431161.2024.2320180.

[46] D. Salido-Monzú and A. Wieser, "An instrumental basis for multispectral LiDAR with spectrally-resolved distance measurements," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 1121–1126, Jun. 2019, doi: 10.5194/isprs-archives-xlii-2-w13-1121-2019.

[47] Y. Han, D. Salido-Monzú, J. A. Butt, S. Schweizer, and A. Wieser, "A feature selection method for multimodal multispectral LiDAR sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 42–57, Jun. 2024, doi: 10.1016/j.isprsjprs.2024.04.022.

[48] Y. Han, D. Salido-Monzú, and A. Wieser, "Comb-based multispectral LiDAR providing reflectance and distance spectra," *Opt. Exp.*, vol. 30, no. 23, pp. 42362–42375, Nov. 2022, doi: 10.1364/oe.473466.

[49] S. Song et al., "A new waveform decomposition method for multispectral LiDAR," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 40–49, Mar. 2019, doi: 10.1016/j.isprsjprs.2019.01.014.

[50] J. Bai et al., "Toward an advanced method for full-waveform hyperspectral LiDAR data processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5702516, doi: 10.1109/TGRS.2024.3382481.

[51] J. Ilinca, S. Kaasalainen, T. Malkamäki, and T. Hakala, "Improved waveform reconstruction and parameter accuracy retrieval for hyperspectral LiDAR data," *Appl. Opt.*, vol. 58, no. 35, pp. 9624–9633, Dec. 2019, doi: 10.1364/ao.58.009624.

[52] Y. Ran, S. Song, X. Hou, Y. Chen, Z. Chen, and W. Gong, "Multi-echo hyperspectral reflectance extraction method based on full waveform hyperspectral LiDAR," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 43–56, Jan. 2024, doi: 10.1016/j.isprsjprs.2023.11.019.

[53] Y. Xia et al., "A novel waveform decomposition and spectral extraction method for 101-channel hyperspectral LiDAR," *Remote Sens.*, vol. 14, no. 21, p. 5285, Oct. 2022, doi: 10.3390/rs14215285.

[54] J. Bai et al., "Full-waveform hyperspectral LiDAR data decomposition via ranking central locations of natural target echoes (Rclonte) at different wavelengths," *Remote Sens. Environ.*, vol. 310, Aug. 2024, Art. no. 114227, doi: 10.1016/j.rse.2024.114227.

[55] B. Wang et al., "Multichannel interconnection decomposition for hyperspectral LiDAR waveforms detected from over 500 m," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515714, doi: 10.1109/TGRS.2021.3108160.

[56] W. Wagner, A. Ullrich, V. Ducic, T. Melzer, and N. Studnicka, "Gaussian decomposition and calibration of a novel small-footprint full-waveform digitising airborne laser scanner," *ISPRS J. Photogramm. Remote Sens.*, vol. 60, no. 2, pp. 100–112, Apr. 2006, doi: 10.1016/j.isprsjprs.2005.12.001.

[57] M. Azadbakht, C. Fraser, and K. Khoshelham, "A sparsity-based regularization approach for deconvolution of full-waveform airborne LiDAR data," *Remote Sens.*, vol. 8, no. 8, p. 648, Aug. 2016, doi: 10.3390/rs8080648.

[58] J. Fang and Y. Wang, "Decomposition of full-waveform LiDAR data utilizing an adaptive B-spline-based model and particle swarm optimization," *Measurement*, vol. 235, Aug. 2024, Art. no. 115002, doi: 10.1016/j.measurement.2024.115002.

[59] X. Shen, Q.-Q. Li, G. Wu, and J. Zhu, "Decomposition of LiDAR waveforms by B-spline-based modeling," *ISPRS J. Photogramm. Remote Sens.*, vol. 128, pp. 182–191, Jun. 2017, doi: 10.1016/j.isprsjprs.2017.03.006.

[60] X. Hou et al., "Multichannel weak signal extraction based on multispectral LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5702311, doi: 10.1109/TGRS.2024.3371575.

[61] J. Bai, Z. Niu, and L. Wang, "A theoretical demonstration on the independence of distance and incidence angle effects for small-footprint hyperspectral LiDAR: Basic physical concepts," *Remote Sens. Environ.*, vol. 315, Dec. 2024, Art. no. 114452, doi: 10.1016/j.rse.2024.114452.

[62] E. Keogh and C. A. Ratanamahatana, "Exact indexing of dynamic time warping," *Knowl. Inf. Syst.*, vol. 7, no. 3, pp. 358–386, Mar. 2005, doi: 10.1007/s10115-004-0154-9.

[63] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-26, no. 1, pp. 43–49, Feb. 1978, doi: 10.1109/TASSP.1978.1163055.

[64] J. Jing, J. Dauwels, T. Rakthanmanon, E. Keogh, S. S. Cash, and M. B. Westover, "Rapid annotation of interictal epileptiform discharges via template matching under dynamic time warping," *J. Neurosci. Methods*, vol. 274, pp. 179–190, Dec. 2016, doi: 10.1016/j.jneumeth.2016.02.025.

[65] M. I. Khalid, T. N. Alotaiby, S. A. Aldosari, S. A. Alshebeili, M. H. Alhameed, and V. Poghosyan, "Epileptic MEG spikes detection using amplitude thresholding and dynamic time warping," *IEEE Access*, vol. 5, pp. 11658–11667, 2017, doi: 10.1109/ACCESS.2017.2718044.

[66] P. Tormene, T. Giorgino, S. Quaglini, and M. Stefanelli, "Matching incomplete time series with dynamic time warping: An algorithm and an application to post-stroke rehabilitation," *Artif. Intell. Med.*, vol. 45, no. 1, pp. 11–34, Jan. 2009, doi: 10.1016/j.artmed.2008.11.007.

[67] P. Ailliot, J. Bessac, V. Monbet, and F. Pène, "Non-homogeneous hidden Markov-switching models for wind time series," *J. Stat. Planning Inference*, vol. 160, pp. 75–88, May 2015, doi: 10.1016/j.jspi.2014.12.005.

[68] X. Chen, X. Ye, X. Xiong, Y. Zhang, and Y. Li, "Improving the accuracy of wind speed spatial interpolation: A pre-processing algorithm for wind speed dynamic time warping interpolation," *Energy*, vol. 295, May 2024, Art. no. 130876, doi: 10.1016/j.energy.2024.130876.

[69] D. J. Dürrenmatt, D. Del Giudice, and J. Rieckermann, "Dynamic time warping improves sewer flow monitoring," *Water Res.*, vol. 47, no. 11, pp. 3803–3816, Jul. 2013, doi: 10.1016/j.watres.2013.03.051.

[70] J. Ares, J. A. Lara, D. Lizcano, and S. Suárez, "A soft computing framework for classifying time series based on fuzzy sets of events," *Inf. Sci.*, vol. 330, pp. 125–144, Feb. 2016, doi: 10.1016/j.ins.2015.10.014.

[71] Y.-S. Jeong, M. K. Jeong, and O. A. Omitaomu, "Weighted dynamic time warping for time series classification," *Pattern Recognit.*, vol. 44, no. 9, pp. 2231–2240, Sep. 2011, doi: 10.1016/j.patcog.2010.09.022.

[72] R. J. Kate, "Using dynamic time warping distances as features for improved time series classification," *Data Mining Knowl. Discovery*, vol. 30, no. 2, pp. 283–312, Mar. 2016, doi: 10.1007/s10618-015-0418-x.

[73] V. Maus, G. Câmara, R. Cartaxo, A. Sanchez, F. M. Ramos, and G. R. de Queiroz, "A time-weighted dynamic time warping method for land-use and land-cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3729–3739, Aug. 2016, doi: 10.1109/JSTARS.2016.2517118.

[74] M. Belgiu and O. Csillik, "Sentinel-2 cropland mapping using pixel-based and object-based time-weighted dynamic time warping analysis," *Remote Sens. Environ.*, vol. 204, pp. 509–523, Jan. 2018, doi: 10.1016/j.rse.2017.10.005.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.

[75] O. Csillik, M. Belgiu, G. Asner, and M. Kelly, "Object-based time-constrained dynamic time warping classification of crops using Sentinel-2," *Remote Sens.*, vol. 11, no. 10, p. 1257, May 2019, doi: 10.3390/rs11101257.
[76] Q. Li, S. Ural, J. Anderson, and J. Shan, "A fuzzy mean-shift approach to LiDAR waveform decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7112–7121, Dec. 2016, doi: 10.1109/TGRS.2016.2596105.
[77] G. Mountrakis and Y. Li, "A linearly approximated iterative Gaussian decomposition method for waveform LiDAR processing," *ISPRS J. Photogramm. Remote Sens.*, vol. 129, pp. 200–211, Jul. 2017, doi: 10.1016/j.isprsjprs.2017.05.009.
[78] T. Zhou and S. C. Popescu, "Bayesian decomposition of full waveform LiDAR data with uncertainty analysis," *Remote Sens. Environ.*, vol. 200, pp. 43–62, Oct. 2017, doi: 10.1016/j.rse.2017.08.012.
[79] M. Bruggisser, A. Roncat, M. E. Schaepman, and F. Morsdorf, "Retrieval of higher order statistical moments from full-waveform LiDAR data for tree species classification," *Remote Sens. Environ.*, vol. 196, pp. 28–41, Jul. 2017, doi: 10.1016/j.rse.2017.04.025.
[80] T. Giorgino, "Computing and visualizing dynamic time warping alignments inR: ThedtwPackage," *J. Stat. Softw.*, vol. 31, no. 7, pp. 1–24, Aug. 2009, doi: 10.18637/jss.v031.i07.
[81] S. Salvador and P. Chan, "Toward accurate dynamic time warping in linear time and space," *Intell. Data Anal.*, vol. 11, no. 5, pp. 561–580, Oct. 2007, doi: 10.3233/ida-2007-11508.
[82] C. Bork, K. Ng, Y. Liu, A. Yee, and M. Pohlscheidt, "Chromatographic peak alignment using derivative dynamic time warping," *Biotechnol. Prog.*, vol. 29, no. 2, pp. 394–402, Mar. 2013, doi: 10.1002/btpr.1680.
[83] R. R. Lindner et al., "Autonomous Gaussian decomposition," *Astronomical J.*, vol. 149, no. 4, p. 138, Mar. 2015, doi: 10.1088/0004-6256/149/4/138.
[84] D. W. Marquardt, "An algorithm for least-squares estimation of nonlinear parameters," *J. Soc. Ind. Appl. Math.*, vol. 11, no. 2, pp. 431–441, Jun. 1963, doi: 10.1137/0111030.
[85] H. Dan, N. Yamashita, and M. Fukushima, "Convergence properties of the inexact Levenberg–Marquardt method under local error bound conditions," *Optim. Methods Softw.*, vol. 17, no. 4, pp. 605–626, Jan. 2002, doi: 10.1080/1055678021000049345.
[86] C. Kanzow, N. Yamashita, and M. Fukushima, "Levenberg–Marquardt methods with strong local convergence properties for solving nonlinear equations with convex constraints," *J. Comput. Appl. Math.*, vol. 172, no. 2, pp. 375–397, Dec. 2004, doi: 10.1016/j.cam.2004.02.013.
[87] G. Liu and J. Ke, "End-to-end full-waveform echo decomposition based on self-attention classification and U-Net decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7978–7987, 2022, doi: 10.1109/JSTARS.2022.3203130.

Yishuo Hao (Student Member, IEEE) received the B.S. degree from China University of Mining and Technology, Beijing, China, in 2023. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Remote Sensing Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include global ecosystem changes, and hyperspectral light detection and ranging (LiDAR) data processing and application.

Zheng Niu received the M.S. degree in geography information system (GIS) and remote sensing from Peking University, Beijing, China, in 1991, and the Ph.D. degree in physical geography from the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing, in 1996.

He is currently a Professor of carbon cycle remote sensing study with the Aerospace Information Research Institute, Chinese Academy of Sciences. His expertise is in the fields of remote sensing related to the global change ecosystems.

Wang Li received the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, China, in 2015.

He is currently an Associate Professor with the Aerospace Information Research Institute, CAS. His main research interests include how to use multisource remote sensing technology to study ecosystem structure and functioning as well as their associated environmental drivers.

Jie Bai received the B.S. degree in remote sensing science and technology from Nanjing University of Information Science and Technology, Nanjing, China, in 2019, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2024.

He is conducting post-doctoral research at the University of Electronic Science and Technology of China, Chengdu, China. His main expertise is centered on quantitative remote sensing, global ecology evaluation, land and vegetation parameter retrieval, and LiDAR/hyperspectral LiDAR basis and application.

Li Wang (Member, IEEE) received the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, China, in 2008.

He is currently a Professor with the Aerospace Information Research Institute, CAS. He has authored or co-authored more than 100 peer-reviewed articles. His research interests include the fields of land use and land cover change, with a focus on land cover and land use change (LCLUC), terrestrial carbon cycle, carbon disturbance, and socioecological consequences of LCLUC.

Kaiyi Bi received the Ph.D. degree in GIS and remote sensing from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2022.

She is currently an Assistant Research Fellow with the Aerospace Information Research Institute, Chinese Academy of Sciences. Her research interests include hyperspectral LiDAR hardware integration and ecological applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:03:08 UTC from IEEE Xplore. Restrictions apply.